Timetables with CP-SAT

Optimization · Level 3 · Lesson 4



In this lesson you will

- Model a club timetable in CP-SAT with all-different rules.
- Explain propagation and search, and what INFEASIBLE proves.
- Add a fairness rule and judge the result.

Key ideas

- A variable per choice, a line per rule
- Cross out, then guess
- INFEASIBLE: a proof
- OPTIMAL: only your rules
- Who gets the bad slot?

Activity: Model a real week

1. List (5 min): Tasks, days, people and rules.
2. Model (8 min): Variables, all-different, fixed days.
3. Run (4 min): Read the status and the timetable.
4. Fairness (3 min): Who gets the worst slot? Add a rule.

No phone? Do it on paper: cross out impossible days for each task, then guess, as the solver does.

Let’s discuss

1. Why doesn’t the solver need to try every plan?
2. What should you do when the answer is INFEASIBLE?
3. Is a timetable that keeps every rule always fair?

Exit ticket

- One rule I would write as a line + one fairness rule for my week.